

# Homer Walke

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## EDUCATION

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### University of California - Berkeley

*PhD in Computer Science*

- Advisor: Professor Sergey Levine

Berkeley, CA

*Anticipated Graduation May 2026*

### Brown University

*BS in Computer Science with Honors*

- Advisors: Professor Michael Littman, Professor Daniel Ritchie

Providence, RI

*May 2021*

## RESEARCH INTERESTS

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- Fundamental generative modeling: diffusion, flow matching, adaptive computation, reasoning.
- Generative modeling for control: vision-language-action models, world modeling.

## RESEARCH EXPERIENCE

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### Graduate Student Researcher

*UC Berkeley, Robotic AI and Learning Lab, Professor Sergey Levine*

September 2021 – Present

### Intern at Physical Intelligence

*Physical Intelligence ([physicalintelligence.company](http://physicalintelligence.company))*

Jan 2025 – Aug 2025

### Undergraduate Researcher

*Brown University, Visual Computing Group, Professor Daniel Ritchie*

June 2020 – May 2021

### Undergraduate Researcher

*Brown University, Robotics Group, Professor Michael Littman*

June 2019 – May 2021

## SELECTED PUBLICATIONS

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### Scaling Cross-Embodied Learning: One Policy for Manipulation, Navigation, Locomotion and Aviation

Ria Doshi\*, Homer Walke\*, Oier Mees, Sudeep Dasari, Sergey Levine.

*CoRL 2024 Oral* | [arXiv:2408.11812](https://arxiv.org/abs/2408.11812).

### Octo: An Open-Source Generalist Robot Policy

Dibya Ghosh\*, Homer Walke\*, Karl Pertsch\*, Kevin Black\*, Oier Mees\*, Sudeep Dasari, Joey Hejna, Charles Xu, Jianlan Luo, Tobias Kreiman, You Liang Tan, Dorsa Sadigh, Chelsea Finn, Sergey Levine.

*RSS 2024* | [arXiv:2405.12213](https://arxiv.org/abs/2405.12213).

### BridgeData V2: A Dataset for Robot Learning at Scale

Homer Walke, Kevin Black, Abraham Lee, Moo Jin Kim, Max Du, Chongyi Zheng, Tony Zhao, Philippe Hansen-Estruch, Quan Vuong, Andre He, Vivek Myers, Kuan Fang, Chelsea Finn, Sergey Levine.

*CoRL 2023* | [arXiv:2308.12952](https://arxiv.org/abs/2308.12952).

## OTHER PUBLICATIONS

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### Knowledge Insulating Vision-Language-Action Models: Train Fast, Run Fast, Generalize Better

Danny Driess, Jost Tobias Springenberg, Brian Ichter, Lili Yu, Adrian Li-Bell, Karl Pertsch, Allen Z. Ren, Homer Walke, Quan Vuong, Lucy Xiaoyang Shi, Sergey Levine

*NeurIPS 2025* | [arXiv:2505.23705](https://arxiv.org/abs/2505.23705).

### $\pi_{0.5}$ : a Vision-Language-Action Model with Open-World Generalization

Physical Intelligence Team

*CoRL 2025* | [arXiv:2504.16054](https://arxiv.org/abs/2504.16054).

**Autonomous Improvement of Instruction Following Skills via Foundation Models**

Zhiyuan Zhou, Pranav Atreya, Abraham Lee, Homer Walke, Oier Mees, Sergey Levine.

*CoRL 2024* | [arXiv:2407.20635](#).

**DROID: A Large-Scale In-The-Wild Robot Manipulation Dataset**

Alexander Khazatsky, Karl Pertsch, et al.

*RSS 2024* | [arXiv:2403.12945](#).

**Evaluating Real-World Robot Manipulation Policies in Simulation**

Xuanlin Li, Kyle Hsu, Jiayuan Gu, Karl Pertsch, Oier Mees, Homer Walke, Chuyuan Fu, Ishikaa Lunawat, Isabel Sieh, Sean Kirmani, Sergey Levine, Jiajun Wu, Chelsea Finn, Hao Su, Quan Vuong, Ted Xiao.

*CoRL 2024* | [arXiv:2405.05941](#).

**Zero-Shot Robotic Manipulation with Pretrained Image-Editing Diffusion Models**

Kevin Black\*, Mitsuhiro Nakamoto\*, Pranav Atreya, Homer Walke, Chelsea Finn, Aviral Kumar, Sergey Levine.

*ICLR 2024* | [arXiv:2310.10639](#).

**Open X-Embodiment: Robotic Learning Datasets and RT-X Models**

Open X-Embodiment Collaboration

*ICRA 2024* | [arXiv:2310.08864](#).

**Goal Representations for Instruction Following: A Semi-Supervised Language Interface to Control**

Vivek Myers\*, Andre He\*, Kuan Fang, Homer Walke, Phillippe Hansen-Estruch, Ching-An Cheng, Mihai Jalobeanu, Andrey Kolobov, Anca Dragan, Sergey Levine.

*CoRL 2023* | [arXiv:2307.00117](#).

**Stabilizing Contrastive RL: Techniques for Offline Goal Reaching**

Chongyi Zheng, Benjamin Eysenbach, Homer Walke, Patrick Yin, Kuan Fang, Ruslan Salakhutdinov, Sergey Levine.

*ICLR 2024 Spotlight* | [arXiv:2306.03346](#).

**Generalization with Lossy Affordances: Leveraging Broad Offline Data for Learning Visuomotor Tasks**

Kuan Fang, Patrick Yin, Ashvin Nair, Homer Walke, Gengchen Yan, Sergey Levine.

*CoRL 2022 Oral* | [arXiv:2210.06601](#).

**Don't Start From Scratch: Leveraging Prior Data to Automate Robotic Reinforcement Learning**

Homer Walke, Jonathan Yang, Albert Yu, Aviral Kumar, Jędrzej Orbik, Avi Singh, Sergey Levine.

*CoRL 2022* | [arXiv:2207.04703](#).

**PLAD: Learning to Infer Shape Programs with Pseudo-Labels and Approximate Distributions**

R. Kenny Jones, Homer Walke, Daniel Ritchie.

*CVPR 2022* | [arXiv:2011.13045](#).

**NeuralLTLf: Learning Linear Temporal Logic Specifications with a Specialized Neural Operator**

Homer Walke, Daniel Ritter, Carl Trimbach, Michael Littman.

*2021* | [arXiv:2111.04147](#).

**Learning Finite Linear Temporal Logic Formulas**

Homer Walke.

*2021* | [Undergraduate Honors Thesis](#)

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**HONORS**

CRA Outstanding Undergraduate Researcher Honorable Mention (2021)

## TEACHING EXPERIENCE

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### **Graduate Student Instructor**

August – December 2022

*UC Berkeley*

- Teaching assistant for CS285 Deep Reinforcement Learning at UC Berkeley.

### **Undergraduate Teaching Assistant**

September – December 2018

*Brown University*

- Teaching assistant for CSCI 0170, one of Brown's introductory computer science courses.

## TECHNICAL SKILLS

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**Languages:** Python, Java, C

**Libraries:** Jax, Tensorflow, PyTorch, NumPy